

PHASE_A

MOTOR CONTROL
PHASE A
AND CANBUS

File: phase_a.kicad_sch

PHASE_B

MOTOR CONTROL
PHASE B

File: phase_b.kicad_sch

PHASE_C

MOTOR CONTROL
PHASE C

File: phase_c.kicad_sch

POWER_USB

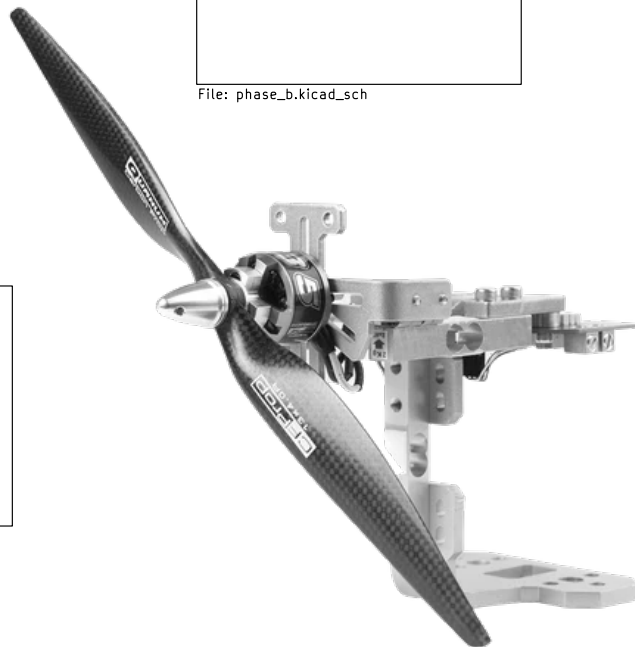
POWER
AND
USB DATA

File: pwr_usb_can.kicad_sch

MCU

STM32G474
MICROCONTROLLER

File: mcu.kicad_sch



DESIGNED FOR 4S, 6S BATTERIES
60A PEAK CURRENT

DRAWN BY: N.PHILIPENKO
GITHUB: /nikphillydev

REVIEWED BY:

Sheet: /
File: ESC_KiCAD.kicad_sch

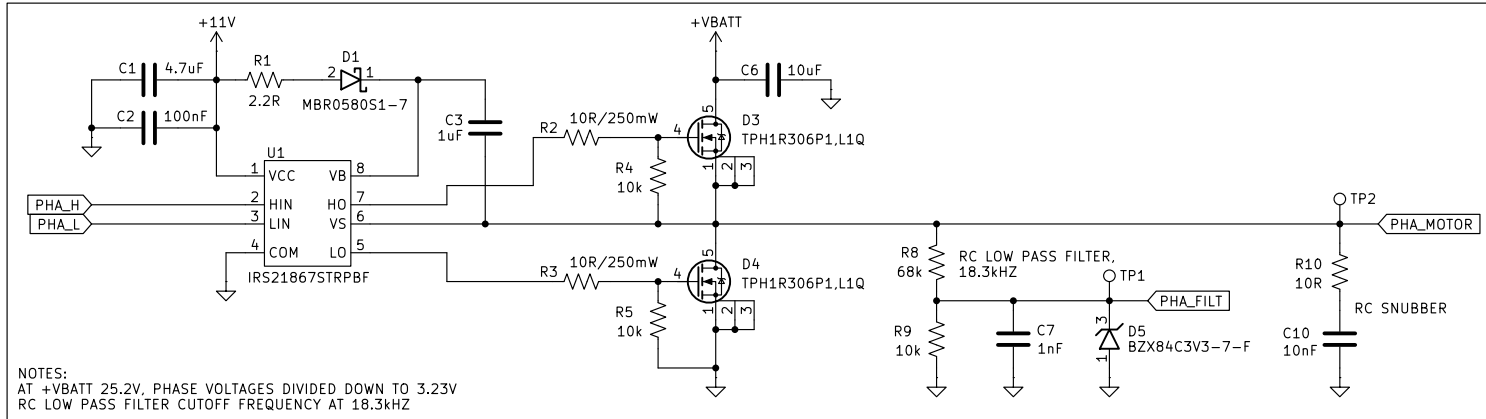
Title: NP1 ELECTRONIC SPEED CONTROLLER

Size: A4 Date: 2025-12-20

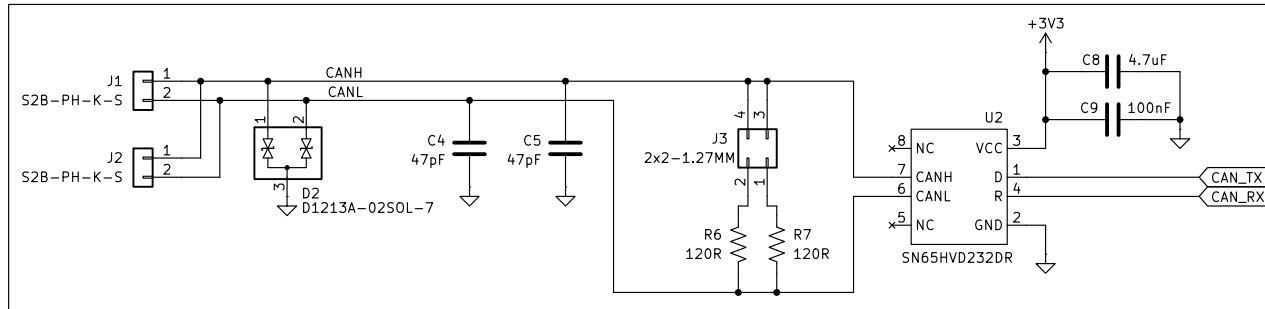
KiCad E.D.A. 8.0.9

Rev: A
Id: 1/6

PHASE A HALF-BRIDGE



ESC CANBUS



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GITHUB: /nikphillydev

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Sheet: /PHASE_A/
File: phase_a.kicad_sch

Title: PHASE A HALF-BRIDGE AND CANBUS

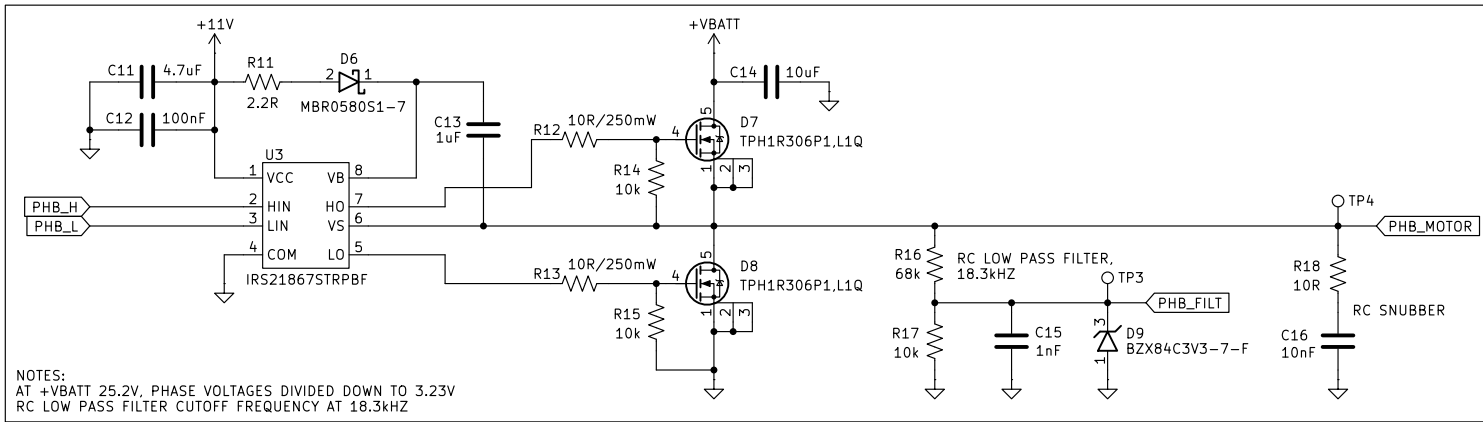
Size: A4 Date: 2025-12-20

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Rev: A

Id: 2/6

PHASE B HALF-BRIDGE



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 GITHUB: /nikphillydev

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Sheet: /PHASE_B/
 File: phase_b.kicad_sch

Title: PHASE B HALF-BRIDGE

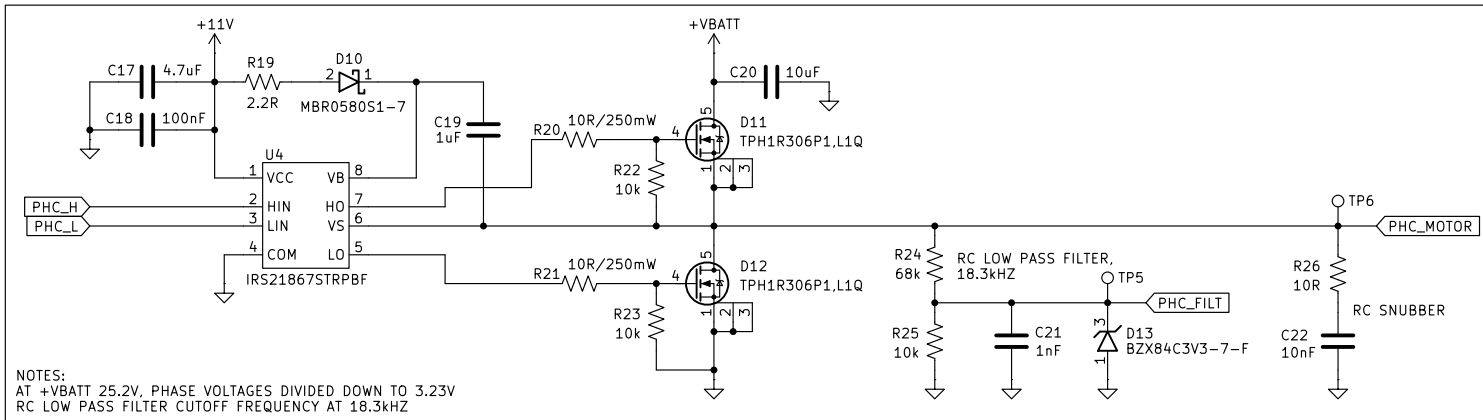
Size: A4 Date: 2025-12-20

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Rev: A

Id: 3/6

PHASE C HALF-BRIDGE



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 GITHUB: /nikphillydev

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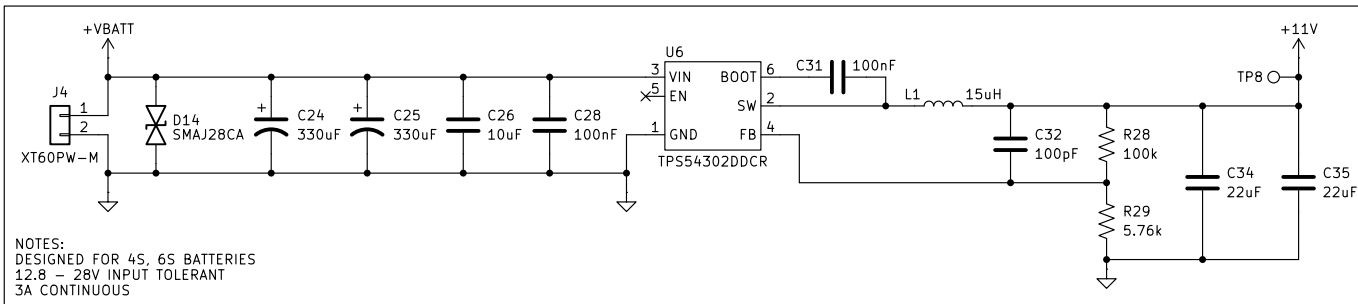
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 File: phase_c.kicad_sch

Title: PHASE C HALF-BRIDGE

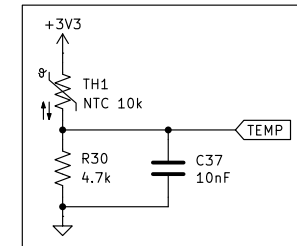
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 KiCad E.D.A. 8.0.9

Rev: A
 Id: 4/6

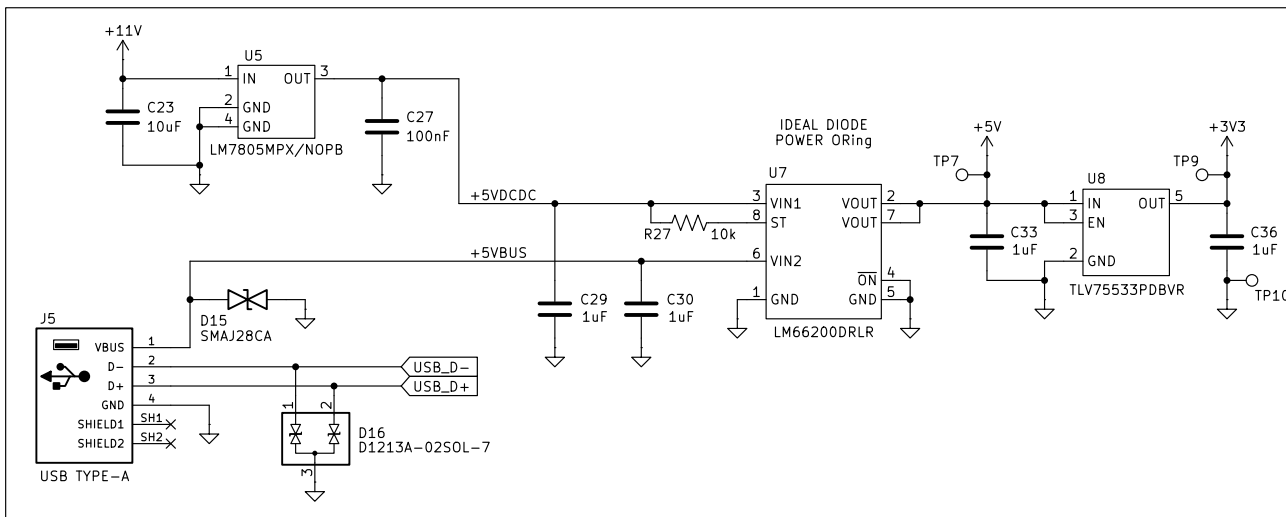
BEC STEP-DOWN CONVERTER



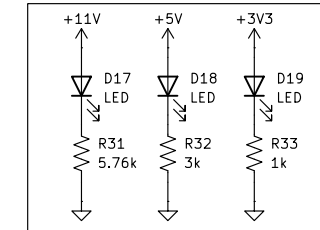
TEMPERATURE MONITORING



LOGIC POWER AND USB DATA



POWER STATUS LEDS



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GITHUB: /nikphillydev

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Sheet: /POWER_USB/
File: pwr_usb_can.kicad_sch

Title: POWER AND USB DATA

Size: A4 Date: 2025-12-20

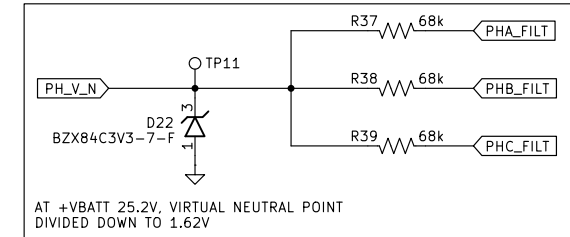
KiCad E.D.A. 8.0.9

Rev: A

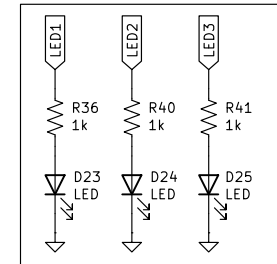
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STM32G474 MCU – NP1 ESC

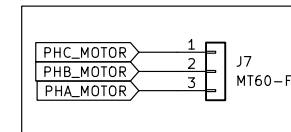
VIRTUAL NEUTRAL POINT (BEMF ZERO-CROSSING DETECTION)



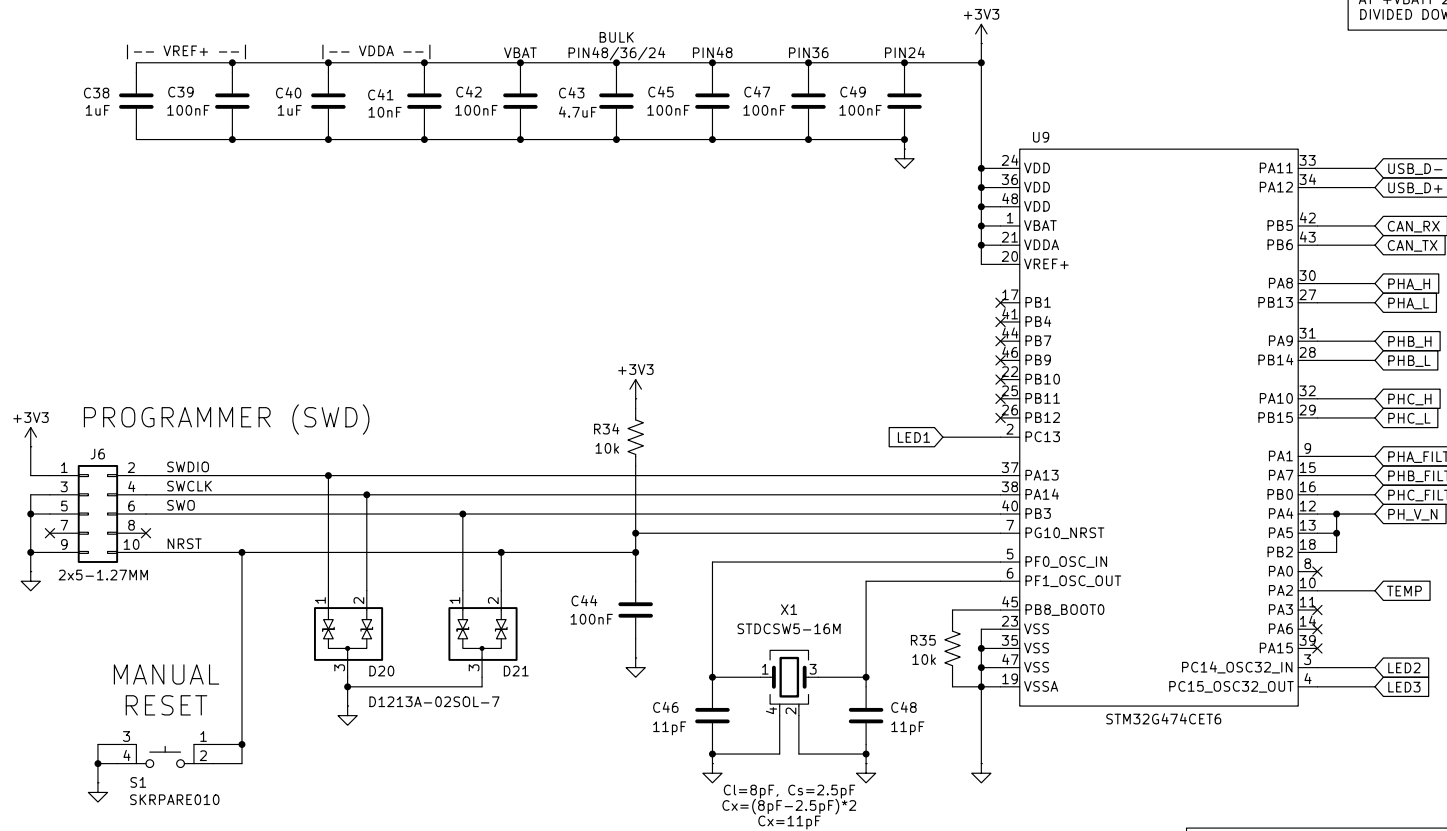
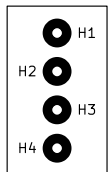
DEBUG LEDS



BLDC MOTOR OUTPUT



MOUNTING HOLES



DRAWN BY: N.PHILIPENKO
GITHUB: /nikphillydev

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Sheet: /MCU/
File: mcu.kicad_sch

Title: MICROCONTROLLER

Size: A4 Date: 2025-12-20

KiCad E.D.A. 8.0.9

Rev: A

Id: 6/6